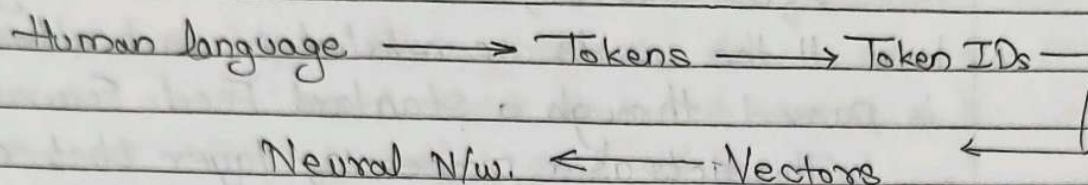


Some basic concepts

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Tokenization

- A NN doesn't naturally understand the concept of a "word".
- It performs mathematical operations on numbers.
- So we need a bridge (Word $\rightarrow$ Nos.)



1) Tokens?

- A unit of text chosen by a tokenizer
Token $\neq$ words
e.g. playing $\Rightarrow$ play + ing $\Rightarrow$ 2 tokens.
- Punctuations can also be a token.
- The exact result depends on the tokenizer.

2) Why not words?

- The tokenizer can't memorize every word, and can even face ~~set~~ newly constructed word it has never seen. So, it tokenizes more often used "subwords" like "ing", "ly", etc.

- tokenizer uses reusable pieces to construct words.

3)

- A word isn't necessarily looked up as a single item, and tokenizer may decompose it.

Conceptually,

Playing $\Rightarrow$ ["Play", "ing"] $\Rightarrow$ [token ids]

for ex, "play" $\rightarrow$ 812

"ing" $\rightarrow$ 451

so, playing $\rightarrow$ [812, 451]

- Exact token ids are tokenizer specific

4) Why do we need token ids?

- Suppose the word is "I love dogs."

The tokenizer might break it into:

["I", "love", "Dogs"]

At this stage, these are still strings.

- ANN doesn't process strings, it needs nos.
- So, tokenizer maintains a vocabulary that looks conceptually like:

Token	ID	Therefore,
I	42	"I love dogs"
love	891	↓
dogs	7124	[I, love, dogs]
cat	2315	↓
car	1098	[42, 891, 7124]

- Taken id is similar to Roll no.

$\rightarrow$ Roll no "1001" doesn't contain any specific info or any meaning.

$\rightarrow$ It is simply an identifier that tells the system:
"Go find the record associated with 1001".

- Similarly, token id "7124" does not mathematically mean animal / pet / canine / furry / 4 legs.

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- It is just an identifier assigned to the token "dogs" in this tokenizer's vocabulary, so, it has no meaning.

5) Where does the meaning enter?

Answer $\rightarrow$ Through learned vector representations.

We have,

[42, 891, 7124]

- Now, suppose the model has an embedding matrix:

$$E \in \mathbb{R}^{V \times d}$$

where, $V \rightarrow$ vocabulary size (no. of distinct token ids the tokenizer knows)

ex. $\rightarrow [0.9, 0.1, 0.7]$
"dog" $d=3$ here

$d \rightarrow$ embedding dimension,

tells how many nos. are used to represent one token.

$E \in \mathbb{R}^{V \times d} \Rightarrow E$ is a $V \times d$ matrix of real nos.

OR E belongs to the set space of $V \times d$ matrices whose entries are real nos. ($\mathbb{R}$).

Suppose, $V=10000$; $d=4$

hence, shape of $E = 10000 \times 4$

Conceptually,

Token	d_1	d_2	d_3	d_4
Token 0	[0.2	0.7	-0.1	0.4]
Token 1	[0.8	-0.2	0.5	0.1]
Token 2	[0.1	-0.4	0.9	-0.3]
Token 42	[0.6	-0.1	0.8	0.2]
...				
Token 891	[0.3	0.7	-0.4	0.9]
...				
Token 7124	[0.2	0.8	0.5	-0.1]
...				

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Token $\rightarrow$ "dogs" $\rightarrow$ piece of language
Token id $\rightarrow$ 7124 $\rightarrow$ address / identifier
Embedding vector $\rightarrow$ $[0.2, 0.8, 0.5, -0.1]$ $\rightarrow$ learned numerical representation

Here, 7124 identifies the token "dogs". It doesn't carry the concept of dogs.

The embedding vector associated with that token is where learned numerical representation begins.

• suppose, dogs $\rightarrow [0.2, 0.8, 0.5, -0.1]$
cats $\rightarrow [0.21, 0.79, 0.48, -0.08]$

$\rightarrow$ they're mathematically similar.

$\rightarrow$ We did not manually tell the model that cats & dogs should have similar vectors.

$\rightarrow$ The model learns useful representation from training

$\rightarrow$ This is the beginning of representation learning.

Now, we can start doing mathematics on language. Before this point, "dogs" was just a string. After representation the model has numbers it can manipulate.

do, basically

"The dog runs" $\rightarrow$ Tokenizer $\rightarrow$ ["The", "dog", "runs"]

Embedding / learned representation $\leftarrow [464, 6213, 1278] \leftarrow$ Token ids $\leftarrow$
 $\rightarrow$ [vector 1, vector 2, vector 3]